Design and Simulation of a Cylindrical Wireless Power Transfer System with Dual Horizontal Resonators for Homecage Monitoring

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***Abstract*—This report presents the design, simulation, and analysis of a wireless power transfer (WPT) system for cylindrical homecage environments used in freely moving animal experiments, targeting a power delivered to load (PDL) of 50 mW for wireless optogenetic implants in mice. We first reproduce the EnerCage-HC2 system, a rectangular homecage WPT design employing segmented copper foil resonators, and validate our HFSS simulation methodology against published results. Building upon this foundation, we propose and iteratively optimize a novel cylindrical homecage WPT system through six design variants. Notably, tilted ring resonators, the cylindrical analog of the EnerCage's tilted rectangular panels, proved entirely ineffective, yielding no improvement over the baseline. The final design (V6), featuring dual horizontal ring resonators, achieves a peak $|S_{21}|$ of -17.6 dB with remarkable spatial homogeneity, only 1.1 dB variation across the lateral axis and 0.7 dB across the vertical axis, significantly outperforming the EnerCage reproduction in spatial uniformity. At this coupling level, a transmitter input power of approximately 3 W would be required to deliver 50 mW to the load, pending impedance matching network optimization and rectifier design. These results demonstrate that cylindrical cage geometries require horizontal rather than tilted resonator topologies, and that dual rings bracketing the receiver height range can provide highly uniform wireless power distribution without requiring animal tracking or coil switching.**

I. Introduction

Behavioral neuroscience research increasingly relies on implantable medical devices (IMDs) to record neural signals and deliver targeted stimulation, such as optogenetic light pulses, in freely moving animal subjects [1]. Traditional hardwired approaches tether the animal to bench-top equipment, restricting natural behavior and confounding experimental results [3]. Battery-powered alternatives allow free movement but require replacement every 2–4 hours, interrupting long-term experiments and posing infection risk when the device is fully implanted [1]. Miniaturized wireless implants, such as the flexible subdermal optogenetic device demonstrated by Shin *et al.* [5], have shown that continuous wireless power delivery can sustain months-long *in vivo* operation, provided the cage infrastructure can supply sufficient and spatially uniform power across the entire experimental volume.

Several wirelessly-powered cage architectures have been developed to address this need. Lee and Jia [1] categorize them into four families: (1) single Tx coil designs, which are simple to drive but suffer from severe coil-size mismatch and low power transfer efficiency (PTE) of 0.56–7.5%; (2) scalable Tx coil arrays, which tile modular coils across the cage floor and activate the nearest one via animal tracking, achieving 5.6–39% PTE at the cost of one power amplifier (PA) per coil; (3) slanted Tx resonator designs, exemplified by the EnerCage system family [3], [8], in which segmented copper foil resonators wrap around the cage and self-localize the magnetic field toward the receiver without tracking or switching, using only a single PA; and (4) omnidirectional systems employing cavity resonators or three-dimensional Rx coil arrangements to handle arbitrary angular misalignments. Among these, the EnerCage-HC2 [3] represents the current state of the art for standard rodent homecages, achieving 14–33% PTE with closed-loop power control (CLPC) and compatibility with standard animal facility racks. However, all systems reviewed by Lee and Jia, as well as subsequent designs, target exclusively rectangular or square cage geometries.

Many behavioral paradigms in neuroscience employ cylindrical enclosures, including the Morris water maze, open field arenas, fear conditioning chambers, and elevated plus-maze center platforms. No wirelessly-powered cage system in the literature has been specifically designed or optimized for cylindrical geometry. This represents a significant gap, because cylindrical symmetry offers distinct advantages for wireless power delivery: a single ring resonator mounted on the cage wall provides uniform angular coverage at a given height, replacing the four separate rectangular panels required in a system like the EnerCage-HC2. Furthermore, the rotational symmetry reduces the spatial optimization problem from three dimensions to two (radial distance and height), simplifying both simulation and design.

In this report, we present the design, simulation, and optimization of a cylindrical WPT system for a 100 mm-diameter mouse cage operating at 13.56 MHz. Our target receiver coil is constrained to $\leq$10 mm diameter to accommodate mice, significantly smaller than the 22 mm Rx used in the EnerCage-HC2 for rats. We first reproduce the EnerCage-HC2 in Ansys HFSS (Section III) to validate our simulation methodology and establish a performance baseline. We then systematically explore six cylindrical cage design variants (Section IV), progressively adding horizontal ring resonators and tuning their parameters. A cross-comparison and discussion of the results follow in Section V, and conclusions are presented in Section VI.

II. Literature Review

A. Overview of Wirelessly-Powered Cage Architectures

Lee and Jia [1] provide a comprehensive survey of wirelessly-powered cage systems, categorizing them into four architectural families based on their transmitter coil configurations. Each family represents a different tradeoff among PTE, spatial homogeneity, system complexity, and scalability.

The simplest approach uses a *single Tx coil* that wraps around or sits beneath the cage. Chow *et al.* [4] employed two series-connected rectangular windings at 6.78 MHz, creating an operating zone of $40 \times 24 \times 4$ cm^3 with PTE of 3.8–7.5%. While straightforward to drive (only one PA required), these designs suffer from severe size mismatch between the large Tx and small Rx coils, yielding low PTE that degrades further as the Rx moves away from the coil center.

The second family employs *scalable Tx coil arrays*, in which identical coils are tiled across the cage floor in overlapping layers. The EnerCage-HC1 system [6] used a three-layer hexagonal planar spiral coil (hex-PSC) array, with the nearest coil to the animal activated via magnetic sensor tracking. This modular approach achieved PTE of 5.6–12.6% with 20 mW PDL under CLPC. However, the architecture requires one PA per driving coil, increasing system complexity and cost proportionally with cage area.

The third family uses *slanted Tx resonators* that wrap around the cage walls, bending and directing the magnetic flux toward the Rx position. The EnerCage-HC system [2] placed four slanted resonators around a standard rat homecage ($46 \times 24 \times 20$ cm^3) and achieved a minimum PTE of 14% everywhere in the cage at the nominal Rx height. This design introduced the *self-localizing effect*: whichever resonator is geometrically closest to the Rx automatically contributes the most power via stronger mutual coupling, eliminating the need for animal tracking or coil switching. The EnerCage-HC2 [3], [9] extended this concept with a fifth Tx resonator and three-dimensional Rx resonators for omnidirectional powering, achieving PTE of 23.6–33.3%. These designs use only a single PA but are dedicated to a specific cage geometry and are not easily scalable.

The fourth family addresses *omnidirectional power transmission*. Cavity resonator designs operating at 346.6 MHz [1] generate rotational H-fields but suffer from blind spots at the center and corners. Triple orthogonal Tx/Rx coil configurations and crossed-dipole coils with DQ rotating magnetic fields can achieve truly orientation-insensitive powering, but their Rx structures are bulky and impractical for miniaturized implants.

Across all four categories, the systems surveyed by Lee and Jia, as well as subsequent designs, exclusively target rectangular or square cage geometries. No wirelessly-powered cage system in the literature has been designed for cylindrical enclosures, despite their widespread use in behavioral neuroscience.

B. EnerCage-HC2: The Benchmark System

The EnerCage-HC2 [3] is the system we reproduce in this work, and therefore warrants a detailed description. It implements a resonance-based four-coil inductive link built around a standard rat homecage at the FCC-approved ISM band frequency of $f = 13.56$ MHz. The Tx side consists of a primary coil L1 (3-turn, 13 cm square, AWG 15 magnet wire, $L_1 = 5.46$ µH, $Q_1 = 116$) positioned at the bottom of the cage, and four tilted rectangular resonators L21–L24 made of 25 mm-wide copper foil. Each resonator is segmented into two equal sections linked by a tuning capacitor, a technique borrowed from MRI coil design to maintain a high self-resonance frequency (SRF) and quality factor ($Q_{21\text{–}24} = 160$–166) at 13.56 MHz. The segmentation rule requires each segment to be shorter than $\lambda_{\text{eff}}/10 = 1.3$ m, where $\lambda_{\text{eff}} = 3 \times 10^8/(f\sqrt{\varepsilon_r}) = 13$ m in the polycarbonate cage substrate ($\varepsilon_r = 3$).

On the Rx side, a resonator L3 (3-turn, 22×20 mm, 5 mm copper foil, $Q_3 = 182$) and a load coil L4 (6-turn, AWG 26, $Q_4 = 138$) are mounted on the headstage at a nominal height of $d = 7$ cm from the cage bottom. The optimal L1 size was derived from the coil sizing rule $A_T = A_R + \pi d^2$, where A_T and A_R are the Tx and Rx coil areas and d is their separation [10].

A key innovation is that the strong mutual coupling among L21–L24 (due to their overlapping placement) creates a single collective resonance frequency for the entire Tx structure. Tuning only one resonator capacitor is sufficient to align all four resonators with the carrier frequency. The self-localizing effect ensures that the resonator(s) closest to the Rx automatically contribute the most power. Combined with a CLPC mechanism implemented via dual BLE links, the system achieves PTE $> 14\%$ everywhere in the cage, with variations of less than 7% at each height level. SAR simulations indicate 0.043 W/kg for the worst-case positioning at 2 W Tx power, well below the 1.6 W/kg IEEE limit.

C. Target Application: Wireless Optogenetic Implants

The end application motivating this work is the wireless powering of miniaturized implantable devices for optogenetic stimulation. Shin *et al.* [5] demonstrated a flexible, fully subdermal wireless optogenetic device weighing only 30 mg with a maximum thickness of 1.3 mm. The device incorporates a planar coil (9.8 mm diameter, 8 turns, 60 µm copper traces, $Q = 22$, $L = 1.8$ µH) resonating at 13.56 MHz via a 23 pF matching capacitor. A Schottky diode rectifies the received RF signal to drive a micro-LED (470 nm, $270 \times 220 \times 50$ µm^3) at the tip of an injectable needle for targeted optical stimulation in the brain.

The device achieved robust *in vivo* operation for over three months without performance degradation, and was successfully demonstrated in reward-seeking behavioral experiments with freely moving mice. The coil's 9.8 mm diameter closely matches the $\leq$10 mm Rx constraint in our cylindrical cage design, making this device class an ideal target application. However, Shin *et al.* powered their implants using a large double-loop antenna external to the cage, which provides limited spatial coverage and no mechanism for homogeneous power delivery across the cage volume, precisely the gap that a dedicated wirelessly-powered cage system would address.

D. Optimal Coil Design for Homecage Monitoring

Chow *et al.* [4] presented a systematic optimization framework for a single-Tx-coil WPT system designed for home-cage monitoring at 6.78 MHz. Their Tx coil comprises two series-connected coaxial rectangular windings, generating an operating zone of $40 \times 24 \times 4$ cm^3 centered at 7 cm above the cage bottom. A π-capacitor matching network on the Rx side ensures that the wireless power delivery is maintained near maximum PDL conditions. The system achieved PTE of 3.8–7.5% and delivered 100 mW to the load.

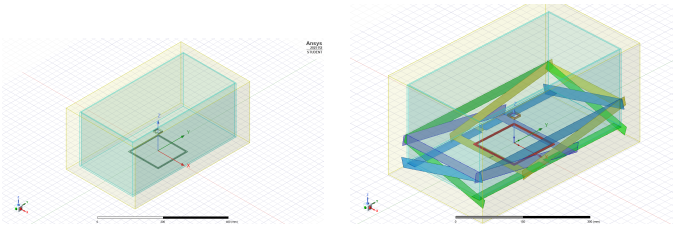
This work highlights the importance of the coil sizing relationship $d_T^2 = d_R^2 + 4d^2$ for optimizing PTE between aligned circular coils separated by distance d, a design rule also employed in the EnerCage-HC2 and adopted in our cylindrical design (Section IV). However, the absence of resonators results in significant PTE degradation at the cage edges, a limitation that the multi-resonator approaches address.

III. EnerCage-HC2 Reproduction

To validate our HFSS simulation methodology before proposing the cylindrical design, we first reproduced the EnerCage-HC2 system in Ansys HFSS 2025 R2 (Student Edition). This section describes the simulation setup, the progressive model refinement, and the final results alongside an honest assessment of the remaining discrepancies with the published data.

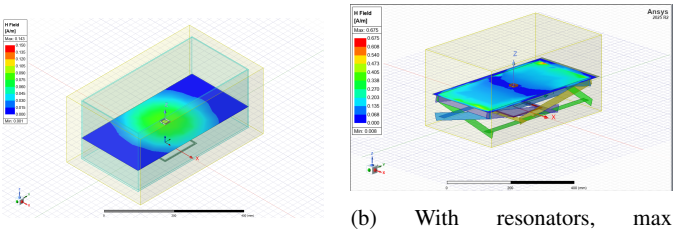
A. Simulation Setup

The HFSS model was constructed following the coil specifications in Table I of [3]. The Tx coil L1 is a 3-turn rectangular spiral with AWG 15 magnet wire (1.45 mm diameter, copper) at the bottom of the cage, with outer dimensions of approximately 129×129 mm^2. Four tilted rectangular resonators L21–L24, each made of 25 mm-wide, 89 µm-thick copper foil, wrap around the cage from the bottom ($Z = -4.45$ mm) up to $Z = 70$ mm. Each resonator is divided into two segments connected by a 130 pF tuning capacitor, matching the published design. On the Rx side, L3 is a 3-turn rectangular



(a) Baseline (L1 + Rx only) (b) With L21–L24 resonators

Fig. 1: EnerCage-HC2 HFSS model geometry: (a) baseline configuration with only L1 and the Rx headstage; (b) full configuration with four tilted segmented resonators wrapping around the cage.



(a) Baseline, max 0.143 A/m (b) With resonators, max 0.675 A/m

Fig. 2: H-field magnitude at $Z = 70$ mm: (a) baseline with L1 only; (b) with L21–L24 resonators. The resonators increase field intensity by $4.7\times$ and distribute it more uniformly across the cage area.

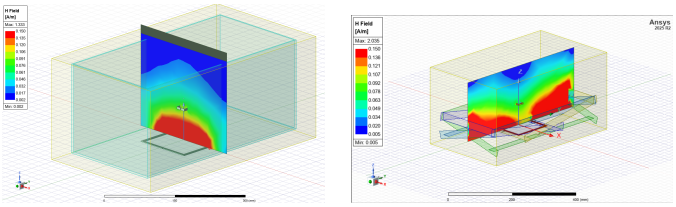
spiral resonator (22×20 mm, 5 mm foil width) and L4 is a 6-turn coil (AWG 26, 0.4 mm diameter), both positioned at the parametric coordinates (`rx_x`, `rx_y`, `rx_z`) with a nominal height of `rx_z` $=$ 70 mm. Port 1 is a 50 Ω lumped port on L1, and Port 2 is a 1000 Ω lumped port on L4. The model is enclosed in a radiation-boundary airbox of $300 \times 500 \times 160$ mm^3.

A significant practical challenge was the solver memory limitation. At 13.56 MHz the free-space wavelength is 22 m, so the default lambda refinement generated an excessively dense airbox mesh (>500k tetrahedra), causing the solver to crash with error code -99 on our 16 GB workstation. Disabling lambda refinement and limiting the adaptive solve to a single pass at 253k tetrahedra resolved this issue, though at the cost of reduced mesh accuracy (estimated 1–2 dB impact on S_{21}).

B. Baseline: L1 + Rx (No Resonators)

We first simulated the system without any resonators, only L1 at the bottom driving L3 and L4 at $Z = 70$ mm, to establish a lower bound on performance. Fig. 4 (dashed blue curve) shows the resulting S_{21}, which peaks at approximately -66 dB near 13.5 MHz. This corresponds to a PTE of roughly 2.5×10^{-5}%, confirming that the direct coupling between the large Tx coil and the small, distant Rx is far too weak for practical power delivery.

The H-field distribution at $Z = 70$ mm (Fig. 2) shows the field concentrated directly above L1 with a maximum of only 0.143 A/m, decaying rapidly toward the cage edges. The XZ-plane field plot reveals that the field drops by more than an order of magnitude between the L1 plane and the Rx height, with negligible field reaching the cage perimeter. A parametric sweep of the Rx along the X-axis at $Z = 70$ mm shows S_{21} varying from -66 dB at the center to below -80 dB at the cage edges, a variation exceeding 15 dB.



(a) Baseline XZ plane (b) With resonators XZ plane

Fig. 3: H-field on the XZ plane: (a) baseline shows rapid field decay above L1; (b) resonators redirect the magnetic flux upward, maintaining strong field at the Rx height.

TABLE I: EnerCage Reproduction: S_{21} Improvement Timeline

Milestone	S_{21} **(dB)**	**PTE (%)**
Initial (wrong C3, C4, C21)	-48.0	0.002
Fixed C21 to 130 pF	-48.0	0.002
C3 tuned to 1200 pF	-19.8	1.0
Port 2 enlarged for contact	-17.7	1.7
C4 location fix + final tune	$\mathbf{-17.4}$	**1.8**
Paper target (center, $d = 7$ cm)	-8.5	14

C. With Segmented Resonators (L21–L24)

Adding the four tilted, segmented resonators dramatically improved both the field distribution and the coupling. The H-field at $Z = 70$ mm increased to a maximum of 0.675 A/m, a $4.7\times$ improvement, and became visibly more uniform across the cage area, with substantial field strength extending to the perimeter. The XZ-plane field plot shows the resonators redirecting the magnetic flux from L1 upward and outward toward the Rx height, consistent with the self-localizing effect described in [3].

The initial S_{21} with resonators was -56.4 dB at 12.6 MHz, a 10 dB improvement over the baseline, but still far from the target. This poor initial performance was traced to several modeling issues documented in detail during our iterative debugging process: (1) the Rx resonator L3 was modeled as a zero-thickness sheet conductor (necessary to avoid mesh failures from its extreme 56:1 aspect ratio), which yielded a lower inductance than the physical copper foil, requiring the tuning capacitor C3 to be increased from the published 40 pF to 1200 pF; (2) one segmentation capacitor (C21) was incorrectly set to 1000 pF instead of 130 pF, splitting the resonance into two weak peaks; and (3) Port 2 was not properly electrically connected to L4 on both sides of its return wire.

D. Final Optimized Design

After systematically resolving the issues above, we performed iterative capacitor tuning at 13.56 MHz. Table I summarizes the progression. The final optimized configuration achieved $S_{21} = -17.4$ dB at 13.6 MHz, representing a cumulative improvement of over 30 dB from the initial model.

Fig. 4 shows the S_{21} vs. frequency for all three configurations overlaid: the baseline (L1 only), the initial resonator model, and the final tuned design. The final design exhibits a clear resonant peak near 13.6 MHz. Fig. 5 shows the spatial sweep of S_{21} along the X-axis at $Z = 70$ mm for the final design. The coupling is strongest near the center (-19.3 dB) and degrades toward the cage edges, reaching -36.5 dB at ± 100 mm.

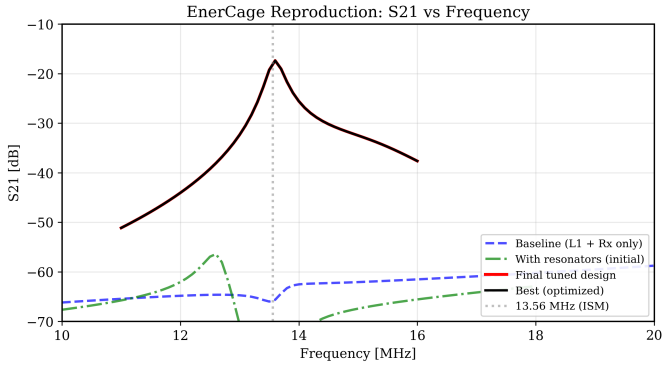


Fig. 4: EnerCage reproduction: S_{21} vs. frequency showing progressive improvement from the baseline (L1 only, dashed blue) through the initial resonator model (dash-dot green) to the final tuned design (solid red and black).

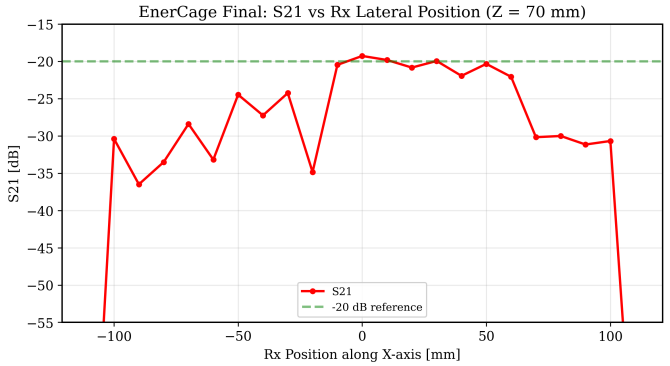


Fig. 5: EnerCage final design: S_{21} vs. Rx lateral position along the X-axis at $Z = 70$ mm. The coupling varies by more than 17 dB across ± 100 mm.

E. Limitations and Discussion

The final reproduced S_{21} of -17.4 dB corresponds to a PTE of approximately 1.8%, compared to the paper's reported 14% (-8.5 dB) at the cage center, a gap of roughly 9 dB. We attribute this discrepancy to several factors.

First, L3 was modeled as a zero-thickness sheet conductor rather than a 3D solid with the physical 89 µm foil thickness, because the extreme aspect ratio caused mesh failures in the student edition of HFSS. This reduces L3's effective inductance, requiring a non-physical capacitor value (C3 = 1200 pF vs. the expected ∼200 pF) and likely degrading the quality factor of the L3 resonance. Second, the adaptive mesh was limited to a single pass due to memory constraints (16 GB RAM), introducing an estimated 1–2 dB error. Third, the Port 2 impedance was set to 1000 Ω, whereas the paper reports optimal load resistances of 45–89 Ω; this mismatch reduces the extracted power. Fourth, the S_{11} at resonance was only -0.46 dB (Fig. 6), indicating that the vast majority of input power is reflected, no impedance matching network was implemented in the simulation.

Despite the absolute PTE gap, the reproduction successfully captures the qualitative behavior of the EnerCage-HC2 system: the resonators produce a clear resonant peak near 13.56 MHz, the H-field distribution at the Rx height becomes dramatically more uniform, and the self-localizing effect is visible in the field plots. The simulation methodology, including the solver settings, meshing strategy, and

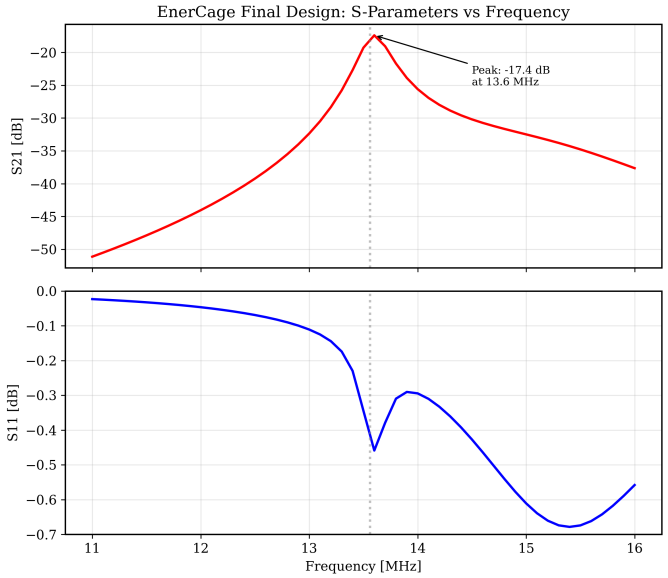


Fig. 6: EnerCage final design: S_{21} (top) and S_{11} (bottom) vs. frequency. The poor S_{11} (> -0.5 dB everywhere) indicates severe impedance mismatch at the input port.

parametric sweep approach, was validated through this process and is directly carried forward to the cylindrical cage design in Section IV.

IV. Cylindrical Homecage Design

A. Design Motivation and Approach

The cylindrical cage design targets freely behaving mice in a 100 mm-diameter, 260 mm-tall polycarbonate enclosure ($\varepsilon_r = 3$) at 13.56 MHz. The Rx coil is constrained to $\leq$10 mm diameter, significantly smaller than the 22 mm Rx in the EnerCage-HC2, which was designed for rats. The nominal Rx height range is $d = 30$–70 mm, corresponding to the mouse head height during walking and rearing.

Cylindrical geometry offers two key advantages over rectangular cages for WPT design. First, rotational symmetry means that a single ring resonator mounted on the cage wall provides uniform angular coverage, replacing the four separate rectangular panels used in the EnerCage-HC2. Second, this symmetry reduces the spatial optimization from a three-dimensional sweep to a two-dimensional one (radial distance r and height z), substantially reducing simulation time.

We adopted a systematic, iterative approach with six design variants of increasing complexity, summarized in Table II. All variants share the same L1 (3-turn circular flat spiral at the cage bottom, AWG 15, outer diameter 100 mm) and L4 (8-turn circular solenoid, AWG 30, 10 mm diameter). The L1 diameter was chosen based on the coil sizing rule $d_T^2 = d_R^2 + 4d^2$: it is optimal for $d \approx 50$ mm, slightly oversized for $d = 30$ mm, and undersized for $d = 70$ mm, making the resonator designs in the later variants critical for covering the upper height range.

B. Variant 1: Baseline

The baseline configuration contains only L1 and L4 with no resonators, forming a simple two-coil inductive link. The L1 coil fills the entire cage bottom (100 mm diameter) and the Rx is centered at $Z = 50$ mm. This variant achieves a peak S_{21} of -30.2 dB at 13.44 MHz, corresponding to a PTE of approximately 0.1%.

TABLE II: Cylindrical Design: Variant Matrix

#	Description	Link	Peak S_{21}
V1	Baseline (L1 + L4 only)	2-coil	-30.2 dB
V2	Horiz. ring resonator (untuned)	3-coil	-13.7 dB
V3	Horiz. ring resonator (tuned)	3-coil	-14.6 dB
V4	Horiz. ring + Rx resonator (L3)	4-coil	-16.9 dB
V5	Tilted ring resonators	3-coil	-30.1 dB
V6	Dual horiz. ring resonators	3-coil	-17.6 dB

While substantially better than the EnerCage baseline (-66 dB), this improvement is expected because the cylindrical cage is much smaller (100 mm vs. 460×240 mm^2) and the Rx is closer to L1 (50 mm vs. 70 mm). Nonetheless, -30 dB is far too low for practical power delivery, confirming the need for resonator enhancement.

C. Variant 2: Single Horizontal Ring Resonator (Untuned)

Variant 2 adds a single horizontal ring resonator on the inner cage wall at $Z = 50$ mm, the middle of the mouse height range. The ring is made of 25 mm-wide copper foil (matching the EnerCage resonator material) with a single turn and a lumped tuning capacitor across a small gap. Unlike the EnerCage's large resonators, the ring circumference ($\pi \times 100 \approx 314$ mm) is far below the segmentation threshold ($\lambda_{\text{eff}}/10 = 1300$ mm), so no segmentation is required.

The untuned resonator immediately improved S_{21} to -13.7 dB, a 16.5 dB gain over the baseline, but the peak frequency shifted to 14.92 MHz, well above the 13.56 MHz target. This result confirmed that the horizontal ring resonator concept is effective for cylindrical cages, but requires capacitor tuning to align the resonance with the operating frequency.

D. Variant 3: Single Horizontal Ring Resonator (Tuned)

Tuning the ring capacitor brought the resonant peak to 13.52 MHz, yielding $S_{21} = -14.6$ dB. A smaller capacitor variant was also tested, producing -14.6 dB at 13.36 MHz, a negligible difference, indicating that the design is robust to small capacitor tolerances.

However, spatial sweep data revealed a significant limitation of the single-resonator approach. Along the lateral axis (rx_x sweep from -40 to 0 mm at $Z = 50$ mm), S_{21} varied from -30.1 dB at center to -36.1 dB at $r = 40$ mm, a 6.0 dB range. More critically, the height sweep (rx_z from 30 to 70 mm at center) showed S_{21} varying from -24.0 dB at $Z = 30$ mm to -35.8 dB at $Z = 70$ mm, an 11.7 dB range. The coupling is strongest close to the resonator height and degrades rapidly with distance, as the single ring creates a narrow band of strong field rather than a volumetric distribution.

E. Variant 4: Horizontal Ring + Rx Resonator (L3)

Variant 4 adds an Rx-side resonator L3 to the V3 configuration, forming a four-coil inductive link. The L3 resonator is coaxial with L4 on the headstage. Counter to expectations, this variant performed slightly worse than V3, achieving $S_{21} = -16.9$ dB at 13.16 MHz. We attribute this to over-coupling: with the Rx already well-coupled to the nearby horizontal ring, adding L3 introduces additional loss through its parasitic resistance without proportionally improving the mutual inductance chain. This finding is consistent with the observation in [7] that three-coil links outperform four-coil links when the coupling is already strong.

F. Variant 5: Tilted Ring Resonators

Inspired by the EnerCage-HC2's tilted rectangular panel resonators, Variant 5 adapts this concept to cylindrical geometry using tilted circular rings. Three rings are spaced at 120° azimuthally around the cage, each tilting from near the bottom ($Z \approx 10$ mm) to above the mouse height range ($Z \approx 90$ mm), with a tilt angle of approximately 53° from horizontal. This arrangement was expected to combine the self-localizing effect of the EnerCage with the angular symmetry of the cylindrical cage.

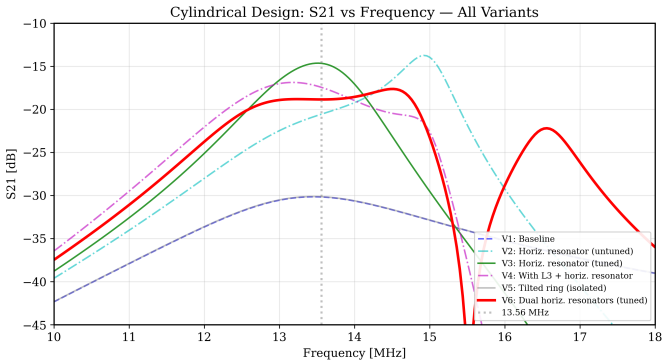


Fig. 7: Cylindrical design: S_{21} vs. frequency for all six variants. V6 (dual horizontal resonators, solid red) achieves comparable peak coupling to V3 (single resonator, green) while providing far superior spatial uniformity (see Fig. 9).

The result was unexpectedly poor: $S_{21} = -30.1$ dB at 13.44 MHz, essentially identical to the baseline. Testing with a single isolated tilted ring confirmed the same performance. The tilted rings couple poorly in this geometry because their orientation means only a small projected area is coplanar with L1 at any given height, resulting in weak mutual coupling. This is a key negative result: the tilted resonator topology that works well for rectangular cages does *not* translate effectively to cylindrical geometries. Horizontal rings, which maintain full area overlap with L1 at a fixed height, are the superior topology for cylindrical cages.

G. Variant 6: Dual Horizontal Ring Resonators

The failure of tilted rings and the height-sensitivity of a single horizontal ring motivated the final design: two horizontal ring resonators at different heights, bracketing the Rx operational range. The lower ring is positioned near $Z = 30$ mm and the upper ring near $Z = 70$ mm, creating constructive field overlap in the region between them.

This configuration achieves a peak S_{21} of -17.6 dB at 14.52 MHz. While the absolute S_{21} is slightly lower than V3's peak (-14.6 dB), the spatial homogeneity is dramatically superior, as detailed in the following subsection. The S_{11} response (Fig. 8) shows multiple resonant dips, at approximately 13, 14.7, and 16 MHz, characteristic of the coupled dual-resonator structure.

H. Spatial Homogeneity Analysis

The spatial homogeneity of V6 is the central result of this work. Fig. 9 compares the S_{21} spatial profiles of V3 and V6 along both the lateral (rx_x) and vertical (rx_z) axes.

Along the lateral axis (Fig. 9a), V6 maintains S_{21} between -17.8 and -18.9 dB across the full sweep from the cage center to $r = 40$ mm, a variation of only **1.1 dB**. In contrast, V3 varies by 6.0 dB over the same range. Along the vertical axis (Fig. 9b), V6 maintains S_{21} between -18.2 dB (at $Z = 40$ mm) and -18.9 dB (at $Z = 70$ mm), a variation of only **0.7 dB**. V3 varies by 11.7 dB over the same height range, with strong coupling only near the single resonator height.

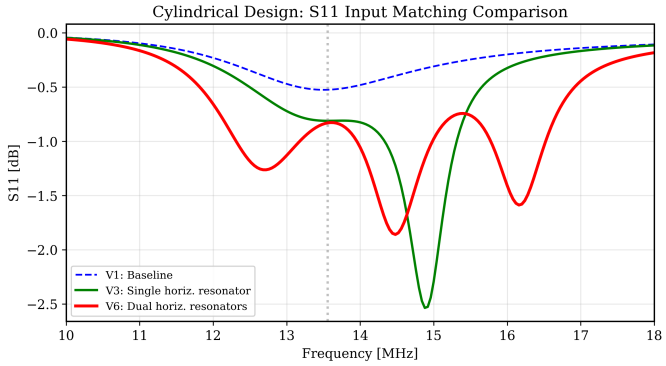
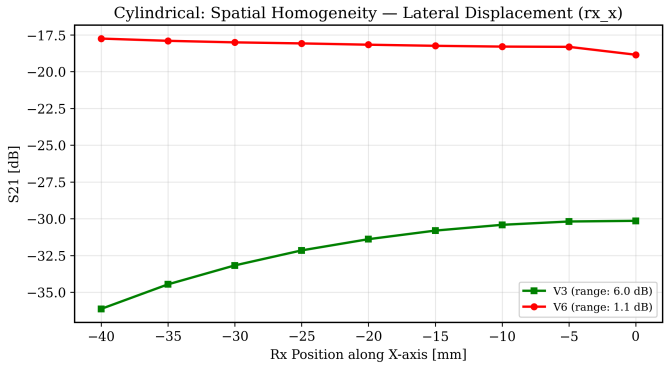
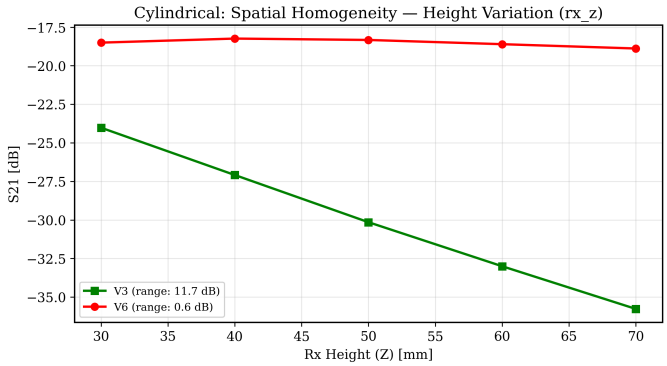


Fig. 8: S_{11} comparison for cylindrical variants V1 (baseline), V3 (single resonator), and V6 (dual resonators). V6 exhibits multiple resonant dips from the coupled dual-resonator structure.



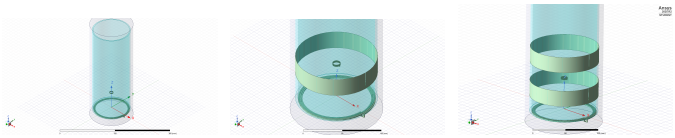
(a) Lateral displacement (rx_x)



(b) Height variation (rx_z)

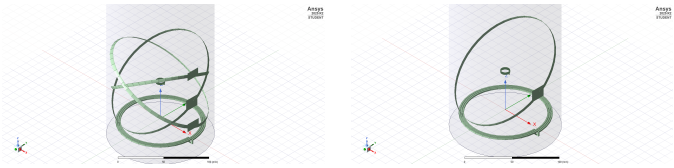
Fig. 9: Spatial homogeneity comparison: V3 (single resonator) vs. V6 (dual resonators). V6 achieves near-uniform S_{21} with only 1.1 dB lateral variation and 0.7 dB vertical variation.

The physical mechanism behind V6's uniformity is the constructive overlap of the magnetic fields generated by the two resonators. Each ring resonator, driven by its mutual coupling to L1, produces a toroidal field pattern centered at its height. When the Rx is near the lower ring ($Z \approx 30$ mm), the lower ring dominates; when the Rx rises to the upper ring ($Z \approx 70$ mm), the upper ring takes over. In the intermediate region ($Z \approx 40$–60 mm), both rings contribute comparably, and their fields add constructively. The result is a nearly flat S_{21} profile across the entire 30–70 mm height range. This is analogous to the self-localizing effect in the EnerCage-HC2, but



(a) V1: Baseline (b) V3: Single ring (c) V6: Dual rings

Fig. 10: HFSS geometry of key cylindrical design variants: (a) V1 baseline with L1 at bottom and Rx above; (b) V3 with a single horizontal ring resonator at $Z = 50$ mm; (c) V6 with dual horizontal ring resonators bracketing the Rx height range.



(a) V5: Tilted rings (3 at 120°) (b) V5: Single tilted ring (isolated)

Fig. 11: V5 tilted ring resonator geometry: (a) three rings at 120° azimuthal spacing; (b) single isolated ring. Both configurations yielded S_{21} identical to the baseline (-30 dB), demonstrating that tilted resonators are ineffective in cylindrical cages.

achieved with horizontal rings rather than tilted panels, and with significantly better uniformity.

The lateral uniformity arises from the cylindrical symmetry of the ring resonators. Unlike the EnerCage's rectangular resonators, which create stronger coupling along certain axes than others, the ring resonator couples equally to the Rx at all azimuthal angles for a given radius. The slight S_{21} increase at larger radii (closer to the ring conductor) is small because the Rx coil is much smaller than the ring, so the coupling geometry changes only gradually with radial position.

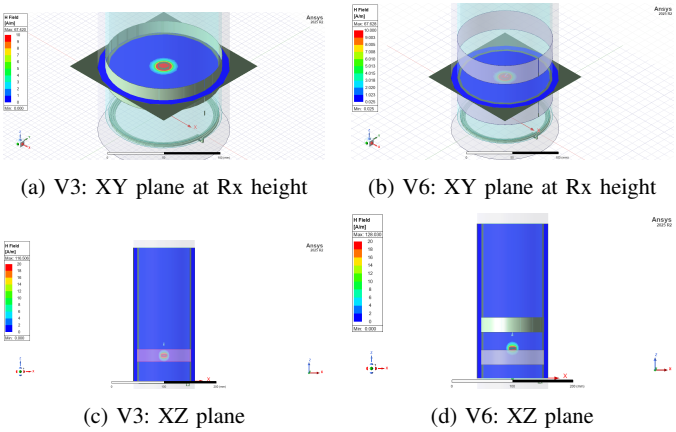
I. Limitations of the Cylindrical Design

Several limitations of the cylindrical cage design and our simulation approach should be noted.

First, the V6 resonant peak occurs at 14.52 MHz rather than the target 13.56 MHz ISM band frequency. While this can likely be corrected through capacitor re-tuning (as demonstrated in the V2→V3 progression), the dual-resonator structure introduces coupled resonances that may make simultaneous tuning of both rings more complex than tuning a single ring. This was not explored in the current work.

Second, the spatial sweeps cover only the radial axis (rx_x from -40 to 0 mm) and vertical axis (rx_z from 30 to 70 mm). The radial sweep is limited to 40 mm, leaving a 10 mm gap before the cage wall at $r = 50$ mm. Performance in this outermost annular region, where the Rx coil approaches the ring resonator conductors, was not characterized and may exhibit anomalous behavior due to near-field effects.

Third, all simulations assume the Rx coil remains perfectly horizontal. In practice, a mouse's head tilts during grooming, rearing, eating, and exploration, introducing angular misalignment between the Rx and Tx. The EnerCage-HC2 addressed this with 3D Rx resonators tilted at 25° on four sides of the headstage [9]. Our cylindrical design uses a simple horizontal Rx coil and does not incorporate angular misalignment compensation. The impact of head tilt on V6's coupling remains unknown.



(a) V3: XY plane at Rx height (b) V6: XY plane at Rx height

(c) V3: XZ plane (d) V6: XZ plane

Fig. 12: H-field comparison between V3 (single ring, left) and V6 (dual rings, right). Top row: XY plane at Rx height. Bottom row: XZ cross-section. V6 shows a broader field band spanning both resonator heights, consistent with the improved spatial homogeneity in S_{21}.

Fourth, the H-field color scales in the current HFSS exports (0–10 A/m for XY cuts, 0–20 A/m for XZ cuts) are too high to reveal the spatial distribution in the region of interest. The field near the ring conductors saturates the scale, while the coupling region between the rings appears uniformly dark blue. Re-exporting with a lower scale (e.g., 0–2 A/m) would better illustrate the uniformity mechanism, but was not completed for this report.

Fifth, no biological tissue model was included in the simulations. The SAR analysis performed for the EnerCage-HC2 [3] showed 0.043 W/kg at 2 W Tx power for a rat-sized tissue model, well within the 1.6 W/kg IEEE limit. Our cylindrical cage is smaller and operates at the same frequency, so SAR is expected to remain within limits, but this must be verified with a mouse-sized tissue phantom before any *in vivo* use.

Finally, the design was not optimized for a specific PDL target. The S_{21} metric characterizes the coupling efficiency of the passive link, but the actual power delivered to the implant depends on the PA output power, matching network losses, rectifier efficiency, and load impedance. A complete system-level analysis would be needed to determine whether V6 can deliver sufficient PDL (typically $>$10 mW for optogenetic applications [5]) across the cage volume.

V. Cross-Comparison and Discussion

A. EnerCage vs. Cylindrical V6

Fig. 13 overlays the spatial S_{21} profiles of the EnerCage reproduction and cylindrical V6 along the lateral axis. The EnerCage achieves slightly better peak coupling at the cage center (-19.3 dB vs. V6's -18.9 dB), but its S_{21} degrades rapidly toward the edges, varying by more than 17 dB across ± 100 mm. In contrast, V6 maintains S_{21} within a 1.1 dB window across ± 40 mm. While the two systems operate in cages of different sizes (the EnerCage's 460×240 mm^2 rectangular cage vs. V6's 100 mm-diameter cylinder), the comparison is meaningful in terms of coverage as a fraction of the cage cross-section: V6 covers the entire cage interior with near-uniform coupling, whereas the EnerCage shows substantial dead zones near the walls.

B. Summary Comparison

Table III summarizes the performance of the key designs. Several observations emerge from this comparison.

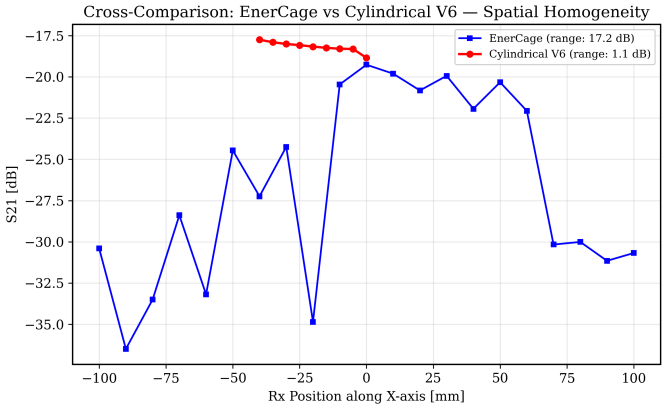


Fig. 13: Cross-comparison: S_{21} vs. Rx lateral position for the EnerCage reproduction (blue) and cylindrical V6 (red). V6 achieves dramatically superior spatial uniformity despite the different cage dimensions.

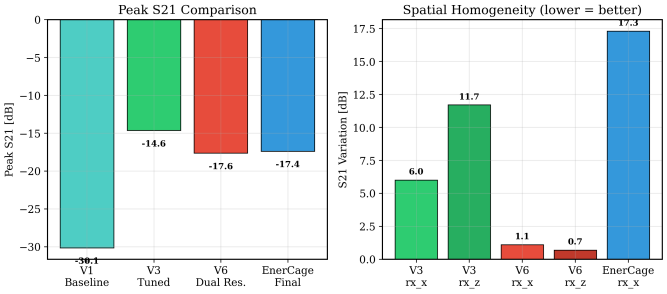


Fig. 14: Summary comparison: (left) peak S_{21} for each design; (right) S_{21} spatial variation in dB (lower is better). V6 achieves comparable peak coupling with far superior homogeneity.

V3 achieves the best peak S_{21} (-14.6 dB) but has the worst spatial uniformity (11.7 dB vertical variation). V6 sacrifices 3 dB in peak coupling relative to V3 but gains near-uniform power delivery across the entire operational volume. For freely moving animal experiments, the minimum PDL anywhere in the cage, not the peak, determines whether the implant remains functional. By this metric, V6 is the clearly superior design: its worst-case S_{21} (-18.9 dB) is better than V3's worst case (-35.8 dB) by 17 dB.

C. Discussion

Input matching. The S_{11} across all designs remains above -3 dB, indicating that the majority of input power is reflected at the source port. This is typical of loosely coupled WPT links simulated without impedance matching networks. In a practical implementation, L-section or π-network matching circuits on both the Tx and Rx sides would improve S_{11} and increase the overall PTE, as demonstrated in [4]. The relative performance ranking among designs would be preserved.

Frequency alignment. V6's resonant peak at 14.52 MHz is approximately 1 MHz above the target ISM frequency of 13.56 MHz. This could be corrected by adjusting the ring resonator tuning capacitors, as demonstrated in the V2→V3 progression where the capacitor shift brought the peak from 14.92 MHz to 13.52 MHz. The dual-resonator structure in V6 may require simultaneous tuning of both ring capacitors.

TABLE III: Performance Comparison of WPT Designs

Metric	V1	V3	V6	EnerCage
Peak $\lvert S_{21}\rvert$ [dB]	-30.2	-14.6	-17.6	-17.4
Freq. at peak [MHz]	13.44	13.52	14.52	13.60
ΔS_{21}, lateral [dB]	—	6.0	1.1	17.3
ΔS_{21}, vertical [dB]	—	11.7	0.7	—
Resonator type	None	1 horiz.	2 horiz.	4 tilted
No. of PAs required	1	1	1	1
Tracking required	No	No	No	No

Peak vs. uniformity tradeoff. The V3 vs. V6 comparison illustrates a fundamental design tradeoff. A single resonator at the nominal Rx height maximizes coupling at that height but creates a narrow effective zone. Distributing the resonator mass across two heights broadens the effective zone at the cost of reducing the peak. For applications where the animal's head height varies (as it does in practice during walking, rearing, eating, and grooming), V6's uniform coverage is more valuable than V3's higher peak.

Advantages of cylindrical geometry. Beyond the performance results, the cylindrical design offers practical benefits: ring resonators are simpler to construct than the four-panel segmented foil arrangement in the EnerCage (one continuous ring vs. four separate panels, each requiring precise slanting and segmentation); the rotational symmetry eliminates the need for Y-axis spatial sweeps, halving the parametric simulation effort; and cylindrical cages are compatible with widely used behavioral paradigms including the Morris water maze, open field test, and fear conditioning chambers.

Limitations. This work has several important limitations. First, all results are from HFSS simulation only, with no experimental prototype validation. The absolute PTE values should be treated as approximate, particularly given the 9 dB discrepancy in the EnerCage reproduction (Section III). However, the relative comparisons between designs, which were all simulated with the same methodology, solver settings, and mesh constraints, are reliable. Second, the spatial sweeps cover only the radial and vertical axes; while cylindrical symmetry guarantees angular uniformity for variants V1–V4 and V6, this was not explicitly verified for V5. Third, no biological tissue model was included, so SAR compliance was not evaluated. Fourth, all simulations assumed a fixed horizontal Rx orientation; angular misalignment from the animal tilting its head was not studied. Finally, the Rx spatial sweep was limited to $r \leq 40$ mm (the cage radius is 50 mm), so performance at the very edge of the cage was not characterized. Finally, the design was not optimized for the 50 mW PDL target. The S_{21} metric characterizes the coupling efficiency of the passive link, but the actual power delivered to the implant depends on the PA output power, matching network losses, rectifier efficiency, and load impedance. At the current $S_{21} = -17.6$ dB (power ratio of 1.74%), approximately 3 W of Tx input power would be needed to deliver 50 mW, assuming ideal rectification and no matching losses. A complete system-level power budget analysis, including matching network design and rectifier characterization, would be required to confirm feasibility.

VI. Conclusion

This report presented the design and HFSS simulation of a wireless power transfer system for cylindrical homecages, motivated by the absence of any WPT cage system optimized for cylindrical geometry in the existing literature.

We first reproduced the EnerCage-HC2 system to validate our simulation methodology. The reproduction captured the qualitative behavior of the published system, resonant coupling near 13.56 MHz, improved H-field uniformity from the tilted resonators, and the self-localizing effect, though the absolute PTE fell short of the published value by approximately 9 dB, attributable to modeling simplifications required by hardware constraints.

We then systematically explored six cylindrical cage design variants. The key findings are:

- A single horizontal ring resonator (V3) improves peak S_{21} by 15.6 dB over the baseline but provides poor height uniformity (11.7 dB variation over $Z = 30$–70 mm).
- Tilted ring resonators (V5), adapted from the EnerCage's rectangular panel approach, are ineffective in cylindrical geometry, yielding no improvement over the baseline.
- Dual horizontal ring resonators (V6) achieve near-uniform wireless power delivery with only 1.1 dB lateral variation and 0.7 dB vertical variation, a $16\times$ improvement in height uniformity over V3.
- Adding an Rx-side resonator (V4) to the single-ring configuration degrades performance, suggesting over-coupling in this compact geometry.

The dual-resonator design (V6) achieves its uniformity through constructive field overlap between the two rings, each of which dominates power delivery when the Rx is near its height. This mechanism is the cylindrical analog of the EnerCage's self-localizing effect, but implemented with horizontal rings rather than tilted panels, and with significantly better spatial homogeneity.

Future work should prioritize experimental validation of the V6 design with a physical prototype, followed by impedance matching network design to improve the currently poor S_{11}, SAR analysis with a tissue phantom model, integration with a closed-loop power control mechanism, and investigation of angular misalignment resilience for the small Rx coil.

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